

Lab 6: Missing Data & Multiple Imputation

Overview

This lab addresses one of the most common challenges in real-world data analysis: **handling missing data**. Simply deleting participants with any missing information shrinks your sample and can introduce serious bias. Instead you will work with **Multiple Imputation (MI)** — creating several plausible complete versions of the data based on what is already known, then checking which conclusions hold across all of them.

Using NHANES data to replicate Williams et al. (2021), you will implement MI inside the complex survey framework from the previous lab, imputing missing confounders and applying the “Multiple Imputation then Deletion” (MID) strategy. Your TA will walk through the worked solution during the session.

This is the workflow you will reuse in **P5** on your own locked paper.

Materials

- **Exercise instructions:** Missing Data Analysis
- **Worked solution:** The R codes

Before you come to lab, install the packages this exercise needs:

```
install.packages(c("mice", "survey"))
```

Your task

1. Follow along as the TA explains the worked solution.
2. Run the R Markdown file yourself, as a group, on your own laptop.
3. Put your **group name and the first names of everyone present** in the `author:` field at the top of the file.
4. Because imputation is random, **set a seed** (for example `set.seed(504)`) so your numbers are reproducible.
5. Add `sessionInfo()` in a final code chunk.
6. Knit to **PDF or HTML** and submit that one file before the lab ends.

No written explanations are required. If you get stuck, ask the TA in the room — that is what the session is for.

Deliverable

- **One file per group:** the knitted PDF or HTML.
- **Due:** end of the lab session (Thu Oct 22).

Grading: Complete / Incomplete

You receive **Complete** if the document knitted without errors, your group name and members are in it, and it was submitted before the lab ended. Only members present in the room receive credit.

Imputation is slow. If the model is still running near the end of the session, reduce the number of imputations (for example `m = 5`) so your document knits in time. You can run the full version later for P5.

Where this lab fits

This lab is one step in a weekly chain: Tuesday's lecture sets up the method, Thursday's lab runs it on fixed teaching data, and your increment applies it to the paper your group locked at M0.

	Document	How this lab connects
This week's lecture	Week 7 — Missing data (MCAR/MAR/MNAR; multiple imputation)	Tuesday covered missingness mechanisms and multiple imputation, and the distinction that matters most: which gaps imputation can address, and which are scope limits to be written up rather than imputed. This lab implements MI inside the survey framework from last week.
Slide deck	Week 7 deck	The same material as slides, with speaker notes. Press S for notes, F for fullscreen.
Your increment	P5 — Missing-data sensitivity	Show your paper's number under missingness stress: a small complete-case / stress-test / MI comparison, and one sentence naming the population your conclusion is actually about.
Worked example	P5 example — its code	A completed P5: the complete-case / stress-test / MI comparison, and the sentence naming the population the conclusion is about.
Milestone	M1 — Replication and interrogation	P5 enters M1 as Missing-data sensitivity . P5 is optional and ungraded with nothing to submit; M1 is where it is assessed.
Builds on	L5 — Survey data analysis	The imputation here runs inside the survey design you declared in L5, so the design and the missingness are handled together rather than one after the other.